CSA1715-ARTIFICIAL INTELLIGENCE

LAB PROGRAM- 19

18-08-2026

19. WRITE A PROLOG PROGRAM FOR STUDENT-TEACHER-SUBJECT-CODE

AIM

To write and execute a Prolog program for maintaining information about students, teachers, subjects, and subject codes. The program demonstrates how relationships between different entities can be represented using facts and rules.

ALGORITHM

1. Start the program.
2. Define facts representing students and their subjects.
3. Define facts representing teachers and the subjects they teach.
4. Define subject codes for each subject.
5. Create rules to establish relationships between students, teachers, subjects, and codes.
6. Query the database to retrieve the required information.
7. Display the result.
8. Stop the program.

PROLOG PROGRAM

```
% Write a Prolog program for Student-Teacher-Subject-Code.


student(rahul, ai).

student(priya, dbms).

student(amit, networks).
```

```
student(neha, ai).

teacher(ramesh, ai).
teacher(suresh, dbms).
teacher(kumar, networks).

subject_code(ai, cs401).
subject_code(dbms, cs402).
subject_code(networks, cs403).

student_teacher(Student, Teacher) :-
    student(Student, Subject),
    teacher(Teacher, Subject).

student_subject_code(Student, Code) :-
    student(Student, Subject),
    subject_code(Subject, Code).
```

SAMPLE INPUT

```
?- student_teacher(rahul, Teacher).
```

SAMPLE OUTPUT

```
Teacher = ramesh.
```

RESULT

Thus, the Student-Teacher-Subject-Code database was successfully created and executed in Prolog. The program correctly retrieved the teacher and subject code associated with a particular student using logical rules.